Certified Strategic Generative AI Professional (CSGAIP)

These objectives focus on the strategic application, integration, and responsible deployment of Generative AI technologies in an enterprise context.

🧠 Domain 1: Foundational Concepts of Generative AI (20%)

This domain tests the candidate's understanding of the core concepts, technologies, and ecosystem of Artificial Intelligence and Generative AI.

- **1.1 AI/ML Foundations**
    - **Define** Artificial Intelligence (AI) and Machine Learning (ML).
    - **Identify** and **compare** the main types of Machine Learning (e.g., supervised, unsupervised, reinforcement).
    - **Differentiate** between key AI categories: Generative AI, Predictive AI, Agentic AI, and others.
    - **Explain** Deep Learning and Natural Language Processing (NLP).
    - **Describe** Generative AI services and the perceptual pillars of AI.
    - **Illustrate** the basic working mechanism of Generative AI. \
- **1.2 Identify Generative AI Technology Stack**
    - **Understand** the comprehensive AI Technology Stack.
    - **Explain** Large Language Models (LLMs) and the Transformer architecture.
    - **Break down** Generative AI model architectures, including Generative Pre-trained Transformers (GPTs).
    - **Describe** the mechanism and benefits of Retrieval-Augmented Generation (RAG). \
- **1.3 Data and Model Lifecycle Management**
    - **Explain** the Data Lifecycle for AI applications.
    - **Understand** the AI Lifecycle and the principles of MLOps.

    - **Relate** the Software Development Lifecycle (SDLC) to AI application development.
    - **Emphasize** the importance of data quality and governance for AI applications.
    - **Describe** the Cloud Computing Shared Responsibility Model in the context of AI services.

🛠️ Domain 2: Prompt Engineering and Model Interaction (15%)

This domain assesses the ability to effectively interact with and optimize Generative AI models using advanced prompting techniques.

- **2.1 Prompt Engineering and Optimization**
    - **Define** Prompt Engineering and Context Engineering.
    - **Compare and contrast** Prompt Engineering vs. Context Engineering.
    - **Demonstrate** how to effectively use the six core components of a prompt:
        - **Role:** Assigning a persona to the AI.
        - **Task/Goal:** Clearly defining the desired outcome.
        - **Context:** Providing relevant background, audience, and purpose.
        - **Format:** Specifying the desired output structure.
        - **Tone/Style:** Guiding the AI's voice.
        - **Constraints/Limitations:** Setting boundaries for the output.
    - **Apply** principles of **Clarity**, **Specificity**, and **Affirmative Directives** in prompt design.
- **2.2 Demonstrate Prompting Strategies**
    - **Apply** the technique of **iterative refinement** to improve model outputs.
    - **Execute Zero-Shot Prompting** to achieve results without explicit examples.
    - **Implement Feedback Loops** to continuously improve prompt effectiveness.
    - **Generate** various types of content using established best practices.

    - **Select** and **utilize** prebuilt Generative Pre-trained Transformers (GPTs) or custom models.

💼 Domain 3: Enterprise Application and Workflow Integration (20%)

This domain focuses on the practical planning, evaluation, and integration of Generative AI solutions into enterprise workflows and systems.

- **3.1 Enterprise Application Planning and Integration**
    - **Identify** general application integration patterns relevant to AI services.
    - **Define** criteria for **evaluation and selection** of Generative AI services and vendors.
    - **Analyze** common AI integration patterns.
    - **Describe** the process of deploying AI services on major cloud platforms (e.g., AWS, GCP).
    - **Identify** and **mitigate** common AI Adoption Challenges.
    - **Apply** AI Adoption Best Practices for successful enterprise implementation.
- **3.2 Use Cases Generative AI for Data-Driven Tasks**
    - **Identify** and **describe** use cases for GenAI in Data Science and Business Intelligence.
    - **Explain** the role of GenAI in Data Augmentation and Synthesis.
    - **Detail** use cases for Code and Content Generation for data analysis tasks.
    - **Analyze** real-world examples of GenAI applied to data analysis and interpretation.

📈 Domain 4: Strategic Leadership and Business Value (Strategic Focus) (30%)

This domain is critical for strategic professionals, covering the development of a business case, vendor strategy, and proposal development for Generative AI initiatives.

- **4.1 Building the Business Case for Generative AI**
    - **Understand** and **apply** the 4 Pillars of AI Strategy.

- Develop a **Value-Driven Roadmap** for AI adoption.
- **Articulate** the Key Value Propositions of Generative AI for the enterprise.
- **Identify** and **measure** Generative AI Value using Strategic Metrics and Uplift Metrics.
- **Construct** a robust **Business Case** for a GenAI project.

- **4.2 Vendor and Technology Strategy**
  - **Differentiate** between Funding Models (CAPEX and OPEX) for AI investments.
  - **Analyze** and **manage** AI Cloud Cost Drivers.
  - **Establish** rigorous **AI Vendor Selection Criteria**.
  - **Formulate** effective **Executive Messaging** to communicate AI strategy.
  - **Manage** the execution and evaluation of Proof of Concepts (PoC).
- **4.3 Strategic AI Proposal Development**
  - **Identify** the essential Components of a Strategic Gen AI Proposal.
  - **Define** the Business Value and Key Performance Indicators (KPIs) for a GenAI project.
  - **Develop** a **Solution Overview & Outcomes** section of a proposal.
  - **Outline** an effective **AI Proposal Workflow** from ideation to approval.

⚖️ Domain 5: Risk, Ethics, and Responsible AI (15%)

This domain validates the candidate's ability to navigate the complex ethical, legal, and risk landscape of deploying Generative AI in the enterprise.

- **5.1 Identify Ethical, Legal and IP Considerations in AI Use**
  - **Define** AI Ethics and its Key AI Principles.
  - **Analyze** the societal impact of AI, including **AI and Job Displacement**.
  - **Identify** and **address** Legal and Professional Accountability issues.
  - **Recognize** common **Sources of AI Bias** and strategies for mitigation.

    - **Explain** the purpose and components of the **NIST AI Risk Management Framework (RMF)**.
    - **Apply** the steps to **implement NIST AI RMF** within an organization.
- **5.2 Implement Best Practices for Responsible AI**
    - **Define** and **explain** Responsible AI Ethics and Governance.
    - **Implement** core Responsible AI Principles.
    - **Understand** and **apply** the Seven C's of Responsible AI.
    - **Describe** the role and function of an **AI Center of Excellence** in driving responsible AI practices.
